Predictive Maintenance ML Model for Delivery Fleet Optimization

Objective

To develop and deploy a predictive maintenance machine learning model within a 12-week timeline that analyzes historical vehicle logs and real-time sensor data.

Planning and Analyzing stages

Milestone	Tasks	Outcome/Deliverables	Estimated Time
Milestone 1	• Define specific vehicle components targeted for failure prediction (e.g., engines, brakes, transmissions). • Identify and secure access to historical maintenance records, performance logs, and real-time IoT sensor streams. •	• finalized technical requirement documentation. • Stakeholders updated.	1- 2 weeks
Milestone 2	• Handle missing values • Exploratory Data Analysis (EDA): •	• Cleaned, unified master dataset ready for modeling. • Stakeholders updated	1-2 weeks

Constructing and Executing stages

Milestone	Tasks	Outcome/Deliverables	Estimated Time
Milestone 3	• Feature Engineering • Model Building • Model Testing	• Trained, validated machine learning model • Stakeholders updated	4 weeks
Milestone 4	• Sharing Results & Insights • Finalize results. • Incorporate feedback	• Final project presentation(visualization) • Excutive summary • result	3 weeks